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TOP STORY

When the stars align: Greenfield Twp. native's experience takes him to Vatican Observatory, NASA

By Ethan Verne For The Valley Advantage
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The Rev. Cyril P. Opeil, right, a Greenfield Twp. native and Lakeland High School graduate, attended the second conference of the Vatican Specola at the Vatican Observatory outside Rome, Italy, in June, where he met Pope Francis. "When you see the Pope, you never expect to shake his hand," Opeil said.

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A Greenfield Twp. Jesuit's research has taken him to the Vatican Observatory — and his studies to the stars.

The Rev. Cyril P. Opeil, S.J., took a yearlong sabbatical from his work as an associate professor of physics at Boston College to accept his invitation to the second conference of the Vatican Specola at the Vatican Observatory in Castel Gandolfo, Italy, outside Rome, where Pope Francis was in attendance.

“It’s really quite extraordinary to meet the pope,” Opeil said. “When you see the pope, you never expect to shake his hand.”

The theme of the scientific conference was “Black Holes, Gravitational Waves and Space-Time Singularities." The June conference was organized in honor of Belgian priest Monsignor Georges Lemaître for his contributions to the scientific community.

A Lakeland High School graduate, Opeil was also one of only a handful of scientists selected to be a part of NASA's OSIRIS-REx mission. The project seeks to learn more about asteroids by visiting, sampling and analyzing the near-Earth asteroid Bennu.

Charting his path

Opeil's life took many different — and often unexpected — turns leading up to these experiences. He graduated from the University of Scranton in 1982 with a degree in electronics engineering.

He was fascinated with how things worked, which led him to work for Westinghouse, where he did computer design for the Air Force. This did not dampen his faith, but rather made it stronger, as he loves to notice the intentionality in everything around us.

"I'm discovering how God has put his thumbprint on creation," Opeil said. "It livens my faith because I'm unraveling the mystery of what is there."

It wasn't until two years later that Opeil made the decision to become a Jesuit. He spent the first two years as a Jesuit novice at Fordham University in the Bronx, New York, studying philosophy and theology.

Due to his blossoming experience in these studies, he was then asked to teach high school physics and chemistry at a high school in Baltimore. This is where Opeil realized his love for teaching, which he carried through his educational career to Boston College, where he earned his Ph.D.

While at the college, Opeil specialized in observing condensed matter, or materials at low temperatures. These materials can have completely different properties at lower temperatures and absolute zero. Opeil was determined to discover how and why these materials react the way they do.

“Science and physics often times have a curious character to them: You discover things when you aren’t looking for them,” Opeil said.

Opeil also worked with scientists at Los Alamos National Laboratory in Los Alamos, New Mexico. There, he was given a device that measures the thermal expansion and contraction of materials, which would prove to be crucial to his studies.

He would also go on to meet Brother Guy Consolmagno, the director of the Vatican Observatory. Consolmagno and other Jesuit

physicists at the observatory have been curating meteorites and extended the opportunity for Opeil to study them.

'Ask the right questions'

Accepting the offer, Opeil began reading up on meteorites, asking the question if they are phyllosilicates, or a type of clay with a very specific structure. This is when he noticed another physicist was scratching his brain trying to figure out why asteroids crack when they are made of dust. As it turns out, Opeil's question was the answer — phyllosilicates.

Using the device he was given at Los Alamos, he ran experiments on the meteorites, observing how they react to changes in temperatures. He discovered something unusual: The meteorites crack at exactly the same temperature.

"You need to recognize when something is unusual so you know that you have something," Opeil said. "You need to know what you're looking for and ask the right questions."

Because these meteorites are phyllosilicates, they crack when exposed to heat, almost like pottery in a way. This went under the radar, as most phyllosilicates have been observed at room temperature.

Opeil is still blown away by the circumstances that had to occur for him to be in this exact position, having the exact right experiences for the job that no one else had.

"There are very few people that actually do this measurement, and it's important to have someone who's set up to do this measurement," Opeil said. "Turns out I'm the only person crazy enough to do this with meteorites."

It is ultimately how these experiences lined up that led him to work with NASA's OSIRIS-REx mission, where he now performs these measurements.

The stars align

Opeil has been working with meteorites for the last nine years. He will continue to do so for the Vatican Observatory and the OSIRIS-REx mission until the end of his sabbatical.

Opeil remarks that, although his life went in all sorts of different directions, he knows that sometimes, the stars will align.

"Serendipity brings you to finding out about the universe," Opeil said. "You notice something unusual, and suddenly you start seeing all of these little connections."